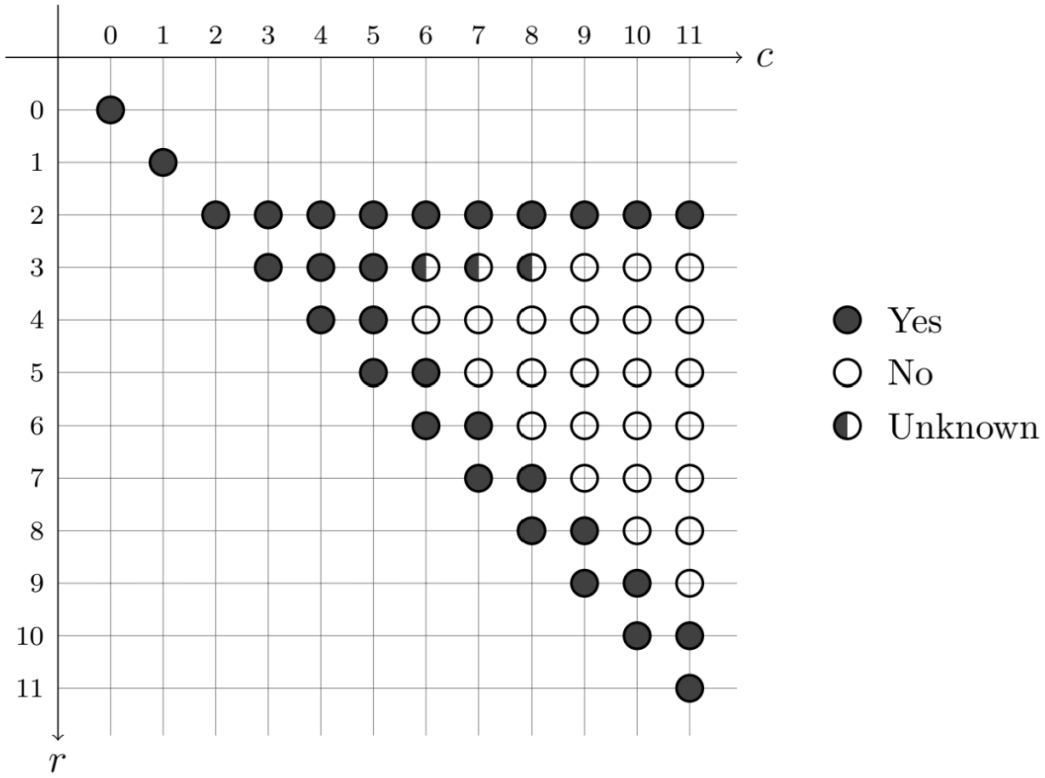
Constructing Non-Koszul Quadratic Gorenstein Algebras II

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Disclaimer

We follow Section 4.2 of "Quadratic Gorenstein Rings and the Koszul Property I" by Matthew Mastroeni, Hal Schenck, and Mike Stillman (2020). **This is not my own work, and the below photo is Figure 2 of this paper.** Thanks to Adam LaClair and Mark Walker for helpful conversations.



The above picture tells us the answer to our main question, "Is every quadratic Gorenstein ring of codimension c and regularity r Koszul?"

So what have we seen in this seminar so far?

Kesavan mentioned alot of these things already, but I honestly don't remember what he actuallly proved for us (see marked up document).

I presented some results last time (draw the picture on the side board and color code what we saw proofs of in the seminar.

Draw the picture and Theorems 2-3 on the side boards before the seminar starts.

Results we discussed previously

Theorem 1. *Let $R = S/I$ be a standard graded (Artinian) k-algebra.*

1. *If R is level, then $\widetilde{R}$ is a standard graded, (Artinian), Gorenstein k-algebra. In this case, one has* $\operatorname{codim}(\widetilde{R}) = \operatorname{codim}(R) + \operatorname{type}(R)$ *and* $\operatorname{reg}(\widetilde{R}) = \operatorname{reg}(R) + 1$. *(The equality comes from the fact that $\widetilde{R}$ is CM and considering the Hilbert series, artinian not required.)*

2. *If R is quadratic and superlevel, then $\widetilde{R}$ is quadratic. (Exploits (4), also without Artinian assumption).*

3. *If R is not Koszul, then $\widetilde{R}$ is not Koszul. (Needs only the Poincare series result of Gulliksen, no Artinian assumption).*

4. $\widetilde{R} \cong S[y_1, \ldots, y_t]/((I) + \mathcal{L} + (y_1, \ldots, y_t)^2)$, *where* $t = \operatorname{type}(R)$ *and*

$$\mathcal{L} = \left(\sum_{i=1}^{t} f_i y_i \mid (f_1, \ldots, f_t) \in \operatorname{Syz}_1^S(\omega_R) \right).$$

(This is all valid without the artinian assumption.)

Definition 1. *By a generic set of g quadrics, we mean a point in the Zariski open subset of* S_2^g.

Theorem 2. *Let* $I \subseteq S = k[x_1, \ldots, x_n]$ *be an ideal generated by* g *generic quadrics, with* $n \geq 4$, $R = S/I$ *artinian, and* g *satisfies the inequalities*

$$\frac{n^2+3n+2}{6} \leq g < \frac{n^2+2n}{4}. \tag{$\star$}$$

Then R *is non-Koszul and superlevel. Consequently,* $\widetilde{R}$ *is a non-Koszul quadratic Gorenstein ring with h-vector*

$$h(\widetilde{R}) = (1, \frac{n^2+3n}{2} - g, \frac{n^2+3n}{2} - g, 1). \tag{$\dagger$}$$

Theorem 3. *(Over a field of characteristic zero) There exist non-Koszul quadratic Gorenstein rings with h-vector* $(1, c, c, 1)$ *for all* $c \geq 9$. *(Notice this answers our main question in the negative for* $r = 3$ *and* $c \geq 9$.*)*

Where were we?

We left off last week attempting to produce more examples by tensoring with other Gorenstein Koszul algebras. We will need a lemma.

Lemma 1. *Let* $R = S/I$ *be a quadratic Gorenstein ring and* B *a superlevel Koszul alegbra. Then* $R' = R \otimes_k B$ *is Koszul (resp. level, superlevel) if and only if* R *is. Further, we have*

$$\begin{aligned} \operatorname{codim} R' &= \operatorname{codim} R + \operatorname{codim} B \\ \operatorname{type} R' &= \operatorname{type} R \cdot \operatorname{type} B \\ \operatorname{reg} R' &= \operatorname{reg} R + \operatorname{reg} B. \end{aligned}$$

Proof Sketch. Here's a blast from the past: R is Koszul iff $\operatorname{reg}_{R'}(R) \leq 1$ ~~and $\operatorname{reg}_{R'}(R) < \infty$.~~

This is because in this case, we can view R as the quotient R'/B since $R' \cong k[\underline{x},\underline{y}]/I+J \cong k[\underline{x}]/I \otimes_k k[\underline{y}]/J$. So this comes from the Transfer of Koszulness talk I gave last semester.

$- \otimes_k -$ is exact [k is a field], so if F is the minimal free res of k over B, then $F \otimes_k R$ is the minimal free resolution of R over R'. Since B is Koszul, $\operatorname{reg}_B(k) = 0$ and so with this observation we obtain $\operatorname{reg}_{R'}(R) = 0$, whence R' is Koszul iff R is Koszul by this blast from the past:

Now write $B = A/J$ for some std grdd poly ring A. Then the other equalities follow from the face that the minimal free res of R' over $S \otimes_k A$ is the tensor product of the minimal free resolutions of R over S and B over A. In particular when R is level, the equalities also follow from the fact that the h-polynomial of R' is the product of h-polynomials of R and B. □

Now we can prove the following Proposition.

Proposition 1. *Over a field of characteristic zero, there exists a non-Koszul quadratic Gorenstein ring of codimension c and regularity r for every $r \geq 6$ and $c \geq r + 3$.*

Proof of Prop 1. Notice that if R is non-Koszul quadratic Gorenstein ring and B is Gorenstein (hence superlevel) Koszul algebra, then $R' = R \otimes_k B$ is again non-Koszul quadratic Gorenstein by Lemma 1.

Now taking an example of Matsuda R (non-Koszul quadratic Gorenstein) with $\operatorname{codim} R = 7$ and $\operatorname{reg} R = 4$ and tensoring with appropraiate Gorenstein Koszul algebras give the results. [This example is a toric ring $K[\mathcal{Q}_{\overline{C_7}}]$, where $\mathcal{Q}_{\overline{C_7}}$ is the so-called stable set polytope of the complement of the cycle graph C_7.]

Specifically, one can take any quadratic Gorenstein ring of $\operatorname{codim} B = 3$ and $\operatorname{reg} R = 2$ (hence the reg ≥ 6 by the Lemma) to be B. By Buchsbaum-Eisenbud's structure theorem (1982) for such rings (quadratic Gorenstein of codimension 3), B must be Koszul. And so $R \otimes_k B$ with these examples produces a negative answer to our Question for $(c, r) = (10, 6)$. We will see an Example that gives us the negative answer for $(c, r) = (9, 3)$ (but actually

this doesn't look like it applies here).

If $X = 3 \times 3$ matrix and $B = k[X]/I_2[X]$ the Gulliksen-Negard resolution shows that $\operatorname{codim} B = 4$ and $\operatorname{reg} B = 2$ so that we can get a negative answer for $(c, r) = (11, 6)$ by taking $B \otimes_k R$ with the same R above. Now using these negative answers, the negative answers in Theorem 3, and tensoring with appropriate complete intersections will give the result. $\square$

Paying the last bit of Proof Debt

We should discuss the proof of Theorem 2, which I promised you all last week.

Proof of Thrm 2. The fact that R is Koszul comes from the computation we did before, where g lying in the appropriate range gives non-Koszulness and the Socle degree 2, from the Froberg-Lofwall, Hochster-Laksov facts. In this computation, we also saw that the h-vector of R must be of the form $(1, n, \binom{n+1}{2} - g) = (1, \operatorname{codim}(R), \operatorname{type}(R))$ [as long as we show superlevel, we will know $\operatorname{type}(R) = h_r$]. Moreover, the form of the h-vector follows from Theorem 1 since $\operatorname{codim}(\widetilde{R}) = \operatorname{codim}(R) + \operatorname{type}(R) = h_1(R) + h_3(R) = \frac{n^2+3n}{2} - g$.

So it is the superlevel that requires some explaining.

Since R is artinian and of socle degree 2 (recall this is a consequence of g falling into the range that it does, it comes from those other results of Froberg-Lofwall, and Hochster-Laksov), it suffices to show that $\beta^S_{n-1,n}(R) = \beta^S_{n,n+1}(R) = 0$, where n of course is the number of variables in S. In fact, due to the upper-semicontinuity of the Betti numbers, it would suffice to show that $\beta^S_{n-1,n}(S/J) = \beta^S_{n,n+1}(S/J) = 0$, for J an initial ideal of I.

Let J denote the initial ideal of I with the degree reverse lexicographic order. So long as J does not contain the smallest $2n-1$ quadratic monomials, we will have $J_2 \subseteq k[x_1, \ldots, x_{n-2}]$. Whence, $\operatorname{pd}_S(S/(J_2)) \leq n - 2$. So the SES

$$0 \to J/(J_2) \to S/(J_2) \to S/J \to 0$$

induces the exact sequence

$$\operatorname{Tor}_i^S(J/(J_2),k)_j \to \operatorname{Tor}_i^S(S/(J_2),k)_j \to \operatorname{Tor}_i^S(S/J,k)_j \to \operatorname{Tor}_{i-1}^S(J/(J_2),k)_j$$

for all i and j. Note that $J/(J_2)$ is generated in degrees at least 3, so $\beta_{i,j}^S(J/(J_2)) = 0$ for all i and $j \leq i+2$. Thus, using the LES in Tor with $i = n-1$ and n gives one the vanishing of the desired betti numbers, i.e. $\beta_{i,i+1}^S(S/J) = \beta_{i,i+1}^S(S/(J_2)) = 0$ for $i = n-1, n$.

We now argue that J can be picked so that it does not contain the smallest $2n-1$ quadratic monomials.

Now since I is generated by generic forms, one may assume that the determinant of the matrix of coefficients of the g largest monomials among all the generators of I is nonzero, which is a Zariski open condition on S_2^g. Thus, after taking appropriate k-linear combinations of the generators of I, one may see that J contains the g largest monomials (in the degree reverse lex order), and these monomials must span J_2 as I and J have the same Hilbert function (likely a standard results about initial ideals). The g largest quadratic monomials are disjoint from the $2n-1$ smallest so long $\binom{n+1}{2} - g \geq 2n-1$. This holds for all $n \geq 7$ via the direct calculation

$$\binom{n+1}{2} - g \geq \binom{n+1}{2} - \left\lfloor \frac{n^2+2n}{4} \right\rfloor = \left\lceil \frac{n^2}{4} \right\rceil \geq 2n-1$$

and via an explicit check when $n = 6$ and $g = 10$.

Now recall that we want $n \geq 4$ and this gives appropriate bounds on g. In particular, when $n = 4$, g must be 5. But the above proof will not work for all the remaining cases, but When $n = 4$, we are covered by the above proposition (which we may have skipped the proof of). When $n = 5$ and $g = 7$, one can check that $(1-t)^5 h_R(t) = (1-t)^5(1+5t+8t^2)$ has no cubic term, whence $\beta_{2,3}^S(R) = 0$.

In the cases that $(n,g) = (5,8), (6,11)$, we claim that $\beta_{3,4}^S(R) = 0$. We may assume that the leading terms of the quadrics generating I are the g largest monomials in degree reverse lex order $x^{\alpha_1}, \ldots, x^{\alpha_g}$. If $\mathcal{B}$ denots the set of exponent vectors of the remaining degree two monomials, then we may assume each quadric has the form $x^{\alpha_i} + \sum_{\beta \in \mathcal{B}} c_{i,\beta} x^\beta$ for some

$c_{i,\beta} \in k$.

By Schreyer's algorithm [whatever that is], one can construct a free resolution F of R (over S) from a Gröbner basis including these quadrics. Now we write $S(-4)^b$ for the number of free summands of $S(-4)$ in F_3 (which doesn't depend on the coefficients of the quadrics), and consider the portion of the differential $\partial_3 : S(-4)^b \to F_2$ (a restriction, since a priori ∂_3 may have a domain with other shifted summands of S). By pruning F one can obtain the minimal free resolution of R, so one obtains $\beta^S_{3,4}(R) = 0$ iff this submatrix of scalars and linear forms splits, which occurs exactly when the scalar part of ∂_3 has rank b. Moreover, the entries of the scalar parts of this submatrix are polynomials in the $c_{i,\beta}$'s so that this determines a Zariski-open condition on S_2^g for the vanishing of $\beta^R_{3,4}(R)$. Thus, $\beta^R_{3,4}(R) = 0$ for a generic set of quadrics if we can show that there is at least one example with this property, and one easily outsources this to M2 by picking g random quadrics. $\square$

We talked a lot about the picture, and did not get to the rest of this, but here it is anyway:

Almost complete intersections (examples)

Recall that we define the a-invariant of R as

$$a(R) = \max\{j \mid H^i_{R_+}(R)_j \neq 0\} = -\min\{j \mid (\omega_R)_j \neq 0\}$$

and

$$a(R) + \dim(R) \leq \operatorname{reg}(R),$$

with equality whenever R is CM. Thus, ω_R is generated at least in degrees $-a(R)$. Finally, R is Gorenstein iff $R(a) \cong \omega_R$ with a the a-invariant (c.f. Section 3.6 in Burns and Herzog).

Proposition 2. *If $R = S/I$ is a quadratic Cohen-Macaulay almost complete intersection with $\operatorname{reg}(R) = 2$ and $\operatorname{height}(I) = 4$, then R is a superlevel algebra but not Koszul.*

Maybe skip this proof for time? The Koszul part is the most interesting but it's just attributed to the fact done by Mastroeni, with the superlevel condition requiring the most work.

Proof. Note that $\operatorname{codim}(R) = \operatorname{height}(I)$ since S is (regular) CM. It is a previous result of Mastroeni [Koszul almost complete intersections, 2018] that any Koszul almost complete intersection of codimension 4 must have regularity 3, so one obtains the non-Koszulness of R rather immediately.

It is left to show that R is superlevel. We reduce to the infinite field artinian case. The conclusion is preserved under flat base change [$\overline{k}$ is a flat k-module] and killing a regular sequence of linear forms on R, so we may assume WLOG that k is infinite and R is Artinian. Hence, we may choose a quadratic complete intersection $L \subseteq I$ with $\operatorname{height} L = \operatorname{height} I = 4$

(so S/J and S/I are both artinian). Put $J = (L : I)$, the ideal directly linked to I by L. Since R is almost ci, J is a Gorenstein ideal (by some linkage results apparently). Now with $a = \operatorname{reg}(S/J)$ we have

$$\begin{aligned}
S/J(a) &\cong \omega_{S/J} && (S/J \text{ CM, so } \operatorname{reg}(S/J) = a(S/J)) \\
&\cong \operatorname{Hom}_S(S/J, \omega_{S/L}) && (L \subseteq J \text{ and } S/J \text{ is Gorenstein}) \\
&= \operatorname{Hom}_S(S/J, S/L(4)) && (S/L \text{ is artinian-ci, so } a(S/L) = 4) \\
&\cong (0 :_{S/L} J/L)(4) && (\text{standard iso}) \\
&= I/L(4). && (\text{containment calculation})
\end{aligned}$$

Since $I/L \cong S/J(a-4)$ (this iso is what we just showed, with adjusted shifts) is generated in degree two, we must have $a = 2$. OTOH, $\omega_R = J/L(4)$ is generated in degrees at least -2 so that J is generated in degree at least 2. Now since S/J is gorenstein with regularity 2, we conclude J must be generated by quadrics, and so ω_R is generated in degree exactly one degree, namely -2, and so R is level.

Now the SES $0 \to J/L \to S/L \to S/J \to 0$ yields

$$\operatorname{Tor}_2^S(S/J, k)_j \to \operatorname{Tor}_1^S(J/L, k)_j \to \operatorname{Tor}_1^S(S/L, k)_j = 0$$

for $j \geq 2$. Since S/J has a Gorenstein linear resolution, we must have $\operatorname{Tor}_2^S(S/J, k)_j = \operatorname{Tor}_1^S(J/L, k)_j = 0$ for $j > 3$. Thus, $\operatorname{Tor}_1^S(\omega_R, k)_j = 0$ for $j > -1$ and ω_R has a linear presentation since it is generated in degree -2. □

Remark 1. *Five generic quadrics in four variables satisfy the conditions of Prop. 2. Thus, one can use this to construct non-Koszul superlevel algebras.*

Example 1. *Consider the quadratic almost complete intersection $R = S/I$ where*

$$I := (x^2, y^2, z^2, w^2, xy + zw) \subseteq k[x, y, z, w] =: S.$$

Notice R is quadratic Artinian (the first four generators of I form a maximal regular sequence in S) and thus CM. Clearly, $\operatorname{height}(I) = 4 = \operatorname{codim}(R)$.

It is not hard to see that $R_+^3 = 0$, *and thus* R *has socle degree* 2 *[i.e.* $R_n = 0$ *for all* $n > 2$*]. The Betti table of* R *is*

	0	*1*	*2*	*3*	*4*
0	*1*	-	-	-	-
1	-	*5*	-	-	-
2	-	-	*15*	*16*	*5*

and thus $\operatorname{reg}(R) = 2$. *Furthermore,* R *is almost ci because* I *is generated by a regular sequence followed by one pesky element. So by the above Prop, we deduce that* R *is superlevel and non-Koszul (the latter also shown by the Betti table). A computation of the h-polynomial shows that* $\operatorname{type}(R) = h_2(R) = \binom{5}{2} - 5 = 5$. *So by Theorem 1,* $\widetilde{R}$ *is a standard graded Artinian, Gorenstein* k*-algebra of codimension* $4 + 5 = 9$ *and regularity* $2 + 1 = 3$. *Further, since* R *is not Koszul, neither is* $\widetilde{R}$*, again by Theorem 1.*

Fact 1. *Any quadratic Cohen-Macaulay almost complete intersection* $R = S/I$ *with* $\operatorname{reg}(R) = n \geq 2$ *and* $\operatorname{codim}(R) = 2n$ *is superlevel but not Koszul. [The argument requires some knowledge of Lefschetz elements and some results of Alexandra and Jason's paper that Kesavan hasn't talked about.]*

Example 2. *One such ring that fits the above criteria and a generalization of the first example is*

$$R = k[x_1, \ldots, x_n, y_1, \ldots, y_n]/(x_i^2, y_i^2 \mid 1 \leq i \leq n) + (\sum_{i=1}^{n} x_i y_i)$$

for all $n \geq 2$.